Module-3

① Explain matplotlib API with suitable Examples?

Matplotlib:- is a python library used to creat graphs and Charts for data visualization. The matplotlib API provide functions and classes to creat, customize, and display plots

type of matplotlib API:-

1. Pyplot API:- simple interface used to creak graphs quickly
2. Obsect-oriented API:- uses figure and Axes obsect to creak and customize plots with more control

Components:-

figure
Axes
Axis
title
Legend.

1. Pyplot API Example:-

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,30,25]
plt.plot(x,y, color='blue', marker='o')
plt.title("Line Graph")
plt.xlabel("x values")
plt.ylabel("y values")
plt.grid(True)
plt.show()
```

A line graph displaying the relationship btw x and y values

2. Object oriented API Example:-

```
import matplotlib.pyplot as plt
x = [1,2,3,4]
y = [5,10,15,20]
fig, ax = plt.subplots()
ax.plot(x,y, marker = 'o')
ax.set_title("Sample Graph")
ax.set_xlabel("x-axis")
ax.set_ylabel("y-axis")
plt.show()
```

Applications:-

⇒ Data analysis and visualization

⇒ comparing value using bar charts

⇒ Showing trends using line graphs

⇒ Displaying distributions using histograms.

⇒ Visualizing scientific and machine learning results

(2) Differentiate Line Plot, Bar plot, Histogram, and scatter plot

A Plot is a graphical representation of data used for analysis and visualization.

Line Plot:-

Purpose:- shows trends over time

Data type:- Continuous or ordered data

Representation:- continous or ordered data points connected by line

x-axis :- Time or value

y-axis :- value

Example:- Temperature over a week

Bar Plot :-

⇒ Compare different categories

⇒ categorical data

⇒ Rectangular bars

⇒ categories

⇒ magnitude

Ex:- marks of students.

Histogram :-

⇒ Shows frequency distribution

⇒ continuous numerical data

⇒ Adjacent bars

⇒ class intervals (bins)

⇒ frequency

Ex:- Distribution of student marks.

Scatter plot :-

⇒ Shows relationship b/w two variable

⇒ Two numerical variable

⇒ Individual data

⇒ One numerical variable

⇒ Another numerical variable

Ex:- Height vs. weight

(3) Explain Seaborn FaceGrid and Categorical Plots.

Seaborn faceGrid :- It is a data visualization tool in Pytho used to creak multiple plots based on different categ- -ories of a dataset. It helps compare data accross groups

Features :-

=> create multiple plots in rows and columns

=> divide data based on categorical variables

=> Helps compare patterns between different groups

=> supports plots such as histograms, scatter plots and line plots

Ex:- import seaborn as sns

```
import matplotlib.pyplot as plt
df = sns.load_dataset("tips")
g = sns.FacetGrid(df, col="sex")
g.map(sns.histplot, "total_bill")
plt.show()
```

Categorical plots :- It is plots are used to visualize and compare numerical data across different categories.

Types of categorical plots:-

Barplot:- Displays the estimated value of a numerical variable for each category

Count plot

Box plot

violin plot

Strip plot

Swarm plot.

Example:-

```
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset("tips")
sns.barplot(data=df, x="day", y="total_bill")
plt.title("Total Bill by day")
plt.show()
```